tinyAM :: CHEATSHEET

01 / DATA + MODEL HYPOTHESES + DIAGNOSTICS | Formula-driven age-structured assessment

1 TIDY THE INPUT

Prepare a validated named list

```
library(tinyAM)
dat <- as_tam_data(cod_obs)
check_tam_data(dat)
names(dat)
```

Tidy schema: one record per year and age

```
head(dat$catch[,    c("year", "age", "obs")])
head(dat$index[,    c("year", "age", "survey", "samp_time", "obs")])
head(dat$weight[,   c("year", "age", "obs")])
head(dat$maturity[, c("year", "age", "obs")])
```

Input tables: **catch** · **index** · **weight** · **maturity** · optional **pop_covariates**. Catchability/observation covariates live in catch or index. Biology and spawning time live in weight/maturity.

2 SPECIFY COMPETING MODEL HYPOTHESES

Start with a deliberately simple baseline

```
base <- fit_tam(
  dat, years = 1983:2024, ages = 2:14,
  N_settings   = list(process = "off", init = "exp"),
  rec_settings = list(process = "iid", mu_form = ~ 1),
  F_settings   = list(process = "iid", mu_form = ~ 1),
  M_settings   = list(process = "off",
                      mu_supplied = ~ I(0.3)),
  catch_settings = list(sd_form = ~ 1),
  index_settings = list(sd_form = ~ 1,
                        q_form = ~ q_block),
  silent = TRUE
)
```

Formula controls: `mu_form` for recruitment and mortality means; `q_form` for survey catchability; `sd_form` for observation/process SDs. For temporal processes use `process = "iid"`, `"rw"` or `"ar1"` as supported by the component.

Change one assumption, refit, and compare

```
F_mono <- update(base,
  F_settings = list(mu_form = ~ mono(age))
)
rec_bh <- update(base,
  rec_settings = list(mu_form = ~ bh(ssb))
)
```

```
q_mono <- update(base,
  index_settings = list(
    q_form = ~ mono(q_block))
)
N_iid <- update(base,
  N_settings = list(process = "iid")
)
```

```
fits <- list(Base = base, F_curve = F_mono,
             BH = rec_bh, Mono_q = q_mono,
             N_process = N_iid)
```

Model structures are hypotheses, not automatically interchangeable. Keep all other settings and inputs fixed where possible; inspect fit quality and identifiability. Ordinary formula examples: `~ 1`, `~ bh(ssb)`, `~ ricker(ssb)`, `~ mono(age)`.

3 VISUALIZE, CHALLENGE AND COMPARE

Compare in an HTML diagnostics dashboard

```
vis_tam(model_list = fits,
        output_file = "models.html",
        open_file = FALSE)
tables <- tidy_tam(model_list = fits)
checks <- lapply(fits, check_tam)
```

Retrospectives: stability of history

```
retro <- lapply(fits, fit_retro, folds = 5)
rho <- lapply(retro, `[[`, "mohns_rho")
```

Compare changes in historical population estimates as recent data are successively removed.

Hindcasts: predictive performance

```
hind <- lapply(fits, fit_hindcast, folds = 5)
rmse <- lapply(hind, `[[`, "hindcast_rmse")
```

Compare one-year-ahead observation predictions. Each function returns the metric; no manual recalculation needed.

tinyAM :: MODEL MATHEMATICS

02 / STATES + PROCESSES + OBSERVATIONS | Compact mathematical reference for the model core

Age-structured states

Years $t = 1, \ldots, T$; ages $a = 1, \ldots, A$; A is the terminal plus group. $N_{t,a}$ is start-of-year abundance; F and M are annual instantaneous mortalities.

$$Z_{t,a} = F_{t,a} + M_{t,a}, \qquad R_t = N_{t,1}.$$

Cohort survival

Expected survivors advance to the next age:

$$\widehat{N}_{t,a} = N_{t-1,a-1} \exp(-Z_{t-1,a-1}), \quad 2 \le a < A.$$

The plus group also retains surviving fish already at age A:

$$\widehat{N}_{t,A} = N_{t-1,A-1} e^{-Z_{t-1,A-1}} + N_{t-1,A} e^{-Z_{t-1,A}}.$$

With an active abundance process:

$$\log N_{t,a} = \log \widehat{N}_{t,a} + \eta^N_{t,a}.$$

Spawning-stock biomass

W: weight; P: mature proportion; s^{spawn}: fraction of the year to spawning. Defaults to zero.

$$SSB_t = \sum_a N_{t,a} W_{t,a} P_{t,a} \, e^{-s^{\text{spawn}}_{t,a} Z_{t,a}}.$$

Shortened plus groups may retain older input biology through effective W and P. Units are not converted automatically.

Initial older-age abundance

`N_settings$init`: `"exp"` uses first-year cohort survivorship, `"free"` estimates older-age states, `"random"` adds IID survivorship deviations. Recruitment at $t = 1$ is a free fixed anchor.

Recruitment dynamics

Default temporal recruitment random walk:

$$\log R_t = \log R_{t-1} + \epsilon^R_t, \quad \epsilon^R_t \sim \mathcal{N}(0, \sigma^2_R).$$

Formula-based means allow environmental covariates and stock-recruitment relationships:

$$\log R_t = \log g(SSB_{t-L}) + X_t \gamma + u_t.$$

$$g_{BH}(S) = \frac{\alpha S}{1 + \beta S},$$

$$g_{Ricker}(S) = \alpha S e^{-\beta S}.$$

The BH/Ricker curves require IID or AR1 recruitment residuals; RW + curve is not supported.

Process variation

For η in N, F or M, process choices encode different hypotheses. Component-specific ages/blocks and SD parameters apply.

$$\begin{aligned} \text{IID}: &\quad \eta_{t,a} \sim \mathcal{N}(0, \sigma^2_a), \\ \text{RW}: &\quad \eta_{t,a} - \eta_{t-1,a} \sim \mathcal{N}(0, \sigma^2_a), \\ \text{AR1}: &\quad \text{Cov}(\eta_{t,a}, \eta_{s,b}) \\ &\quad = \frac{\sigma^2 \, \phi_t^{|t-s|} \phi_a^{|a-b|}}{(1-\phi_t^2)(1-\phi_a^2)}. \end{aligned}$$

F also offers correlated age-specific RW increments with $\Sigma_{ab} = \sigma_a \sigma_b \rho^{|a-b|}$. The RW starting level is unpenalized.

Mortality mean structure

$$\log \mu^F = X_F \beta_F, \qquad \log \mu^M = \log M^{\text{supplied}} + X_M \beta_M.$$

Where fitted, realized log-F/M states depart from these mean surfaces; `mono()` or `logistic()` can describe rising F mean curves under IID/AR1 residuals.

Catch and survey predictions

Baranov catch equation (numbers at age):

$$\widetilde{C}_{t,a} = N_{t,a} \frac{F_{t,a}}{Z_{t,a}} (1 - e^{-Z_{t,a}}).$$

Survey index at within-year sampling time $s_i \in [0, 1]$:

$$\tilde{I}_i = q_i N_{t_i,a_i} e^{-s_i Z_{t_i,a_i}}.$$

Default catchability link: $q_i = \exp(\eta_{q,i})$; optional logit link confines q_i to $(0, 1)$. With `mono(q_block)`, successive increments in η_q are nonnegative.

Observation distributions

Observed positive catch and index values have lognormal likelihood contributions:

$$\log Y_i \sim \mathcal{N}(\log \widetilde{Y}_i, \tau^2_i),$$

$$\tau_i = \tau_i^{\text{supplied}} \exp(X_{sd,i} \beta_{sd}).$$

$\widetilde{Y}$ is a conditional median. Zero/NA observations are treated as missing. Observation SD is distinct from estimation SE.

Likelihood and inference

The joint likelihood combines recruitment, active N/F/M processes, random initial abundance and observation densities:

$$\ell(\theta, \eta) = \ell_R + \ell_N + \ell_F + \ell_M + \ell_{N_0} + \ell_{\text{obs}}.$$

`fit_tam()` fits by RTMB Laplace integration of random effects and `nlminb()` optimization. Reported uncertainty is a local curvature/delta-method approximation.

Assess predictive skill

Retrospective peels assess stability (Mohn's ρ). Hindcasts hold out terminal observations and use one-step-ahead projection errors; `fit_retro()` and `fit_hindcast()` return summary statistics.